Health Discourse in Transition

Semantic Similarity between Italian Parliamentary Parties
before and during COVID-19

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Abstract

This report extends an earlier group project on Italian parliamentary health discourse, shifting the focus from lexical differences between political blocs to semantic similarities across individual party categories. Using ItaParlCorpus, we identify health-related speeches through BERTopic and analyse 2,400 interventions assigned to five source categories: PD, Fratelli d'Italia, Movimento 5 Stelle, Forza Italia and Lega. Contextual speech embeddings are averaged into party-period profiles for 2018–2019, 2020 and 2021–2022. We compare these profiles using cosine distance and hierarchical clustering, assessing uncertainty through debate-date bootstrap resampling. The results reveal changing semantic groupings: an unstable pre-pandemic partition, a stable PD–M5S versus FI–Lega–FdI split in 2020, and a separate FdI branch in 2021–2022. The latter partition recurs in 99.8% of bootstrap samples. An extended health selection preserves the principal partitions, although some pairwise distances change substantially. Equal speaker weighting also preserves the two later-period partitions, but changes the initial one. These findings describe shifts in average semantic content rather than direct political agreement or causal effects of the pandemic. The accompanying GitHub repository documents the data preparation and analysis workflow.

1 Introduction

The COVID-19 pandemic provides a setting in which to examine changes in parliamentary health discourse. Comparing speeches before and during the pandemic allows us to investigate how the content discussed by different political parties evolved, and whether their discourse became more or less similar.

This report extends an earlier group project on Italian parliamentary health discourse, in which the author participated. The original analysis examined lexical differences between political blocs. The present extension shifts the focus to contextual representations of speeches and compares five individual party categories: Partito Democratico (PD), Fratelli d'Italia (FdI), Movimento 5 Stelle (M5S), Forza Italia (FI), and Lega.

The research question is: *How did the semantic similarity of health-related parliamentary discourse across these five party categories evolve before and during the COVID-19 pandemic?* We examine both differences between parties within each period and changes within each party over time.

The analysis combines topic-based speech selection with contextual embeddings. BERTopic identifies a health-related subset of the parliamentary corpus. Speech embeddings are then aggregated into party profiles for three calendar periods: 2018–2019, 2020, and 2021–2022. Cosine distances and hierarchical clustering describe their relative proximity. Debate-date bootstrap resampling assesses uncertainty, while a broader health selection checks sensitivity to the definition of the sample.

The contribution is a descriptive account of changing semantic groupings, supported by a documented computational workflow. Semantic proximity is not interpreted as direct political agreement: speeches may address similar subjects while defending different positions. Likewise,

the temporal comparisons do not identify a causal effect of the pandemic. Party categories are inherited from the source dataset and may differ from parliamentary affiliations at the date of an intervention.

The accompanying GitHub repository provides the preparation notebooks, saved embeddings, and analysis outputs underlying this report.

2 Data and Methods

2.1 Corpus and sample selection

We use ItaParlCorpus version 3.0 [1], drawing on the `camera_2006-2022.csv` file available through Harvard Dataverse. The preparation pipeline selects speeches from legislature XVIII within 2018–2022 and excludes interventions marked as chair contributions. This produces 61,985 records, covering 29 March 2018 to 28 September 2022.

Original texts and metadata are preserved alongside cleaned texts. Cleaning removes known speaker headers and selected transcription directions. A separate Italian lexical representation is produced through tokenization, lemmatization, and stopword filtering. Speeches are eligible for modelling if they contain at least 20 whitespace-separated words and five retained lexical tokens, and are neither empty nor flagged as transcription headings. These criteria retain 49,629 speeches.

BERTopic is fitted to all eligible speeches, retaining all source party categories at this stage. Inspection of topic keywords and speech examples identifies topic 0 as the main health-related selection, containing 3,415 speeches. Restricting this selection to PD, FdI, M5S, FI, and Lega yields the main analytical sample of 2,400 speeches. An extended selection adds topic 14, concerning end-of-life discourse, and contains 2,745 speeches across the five categories. Topic identifiers refer to the saved model run; the selections are approximate rather than validated health classifications.

We retain the party labels supplied by ItaParlCorpus without manual affiliation corrections. These labels may differ from the parliamentary group indicated in an intervention. Accordingly, the analysis compares source party categories rather than independently verified affiliations at each date.

2.2 Speech embeddings and topic identification

Cleaned speeches are encoded using `paraphrase-multilingual-MiniLM-L12-v2`, a multilingual Sentence Transformer producing 384-dimensional vectors [2]. Long speeches are divided into non-overlapping passages within the model's 128-token input limit, including special tokens. Passage embeddings are averaged using their counts of non-special, non-padding tokens as weights. The resulting speech vectors are normalized to unit length.

BERTopic [3] clusters the speech embeddings after UMAP [4] reduces their dimensionality to five components. UMAP uses 15 neighbours, cosine distance, and random seed 42. HDBSCAN [5] uses a minimum cluster size of 100 and `min_samples` set to 10. The lemmatized texts supply the topic keyword representations. The fitted model produces 69 topics, with 20,629 speeches unassigned to a cluster. These unassigned speeches are excluded from the topic-based health selections.

2.3 Party-period profiles and semantic distances

We divide the sample into three calendar periods: 2018–2019, 2020, and 2021–2022. The first and final years have partial coverage, 2020 includes months preceding the emergency, and 2021–2022 represents the later pandemic rather than a post-pandemic period.

Let $\mathbf{x}_i$ denote the unit embedding of speech i. For party category g in period t, we calculate the mean embedding and normalize it:

$$\mathbf{m}_{gt} = \frac{1}{n_{gt}} \sum_{i \in S_{gt}} \mathbf{x}_i, \qquad \mathbf{c}_{gt} = \frac{\mathbf{m}_{gt}}{\|\mathbf{m}_{gt}\|},$$

where S_{gt} contains the group's speeches and n_{gt} is its size. Each speech receives equal weight.

The cosine distance between two profiles is

$$d((g,t),(h,s)) = 1 - \mathbf{c}_{gt}^{\top}\mathbf{c}_{hs}.$$

We calculate all ten between-party distances within each period and all three temporal distances within each party category. Hierarchical clustering uses these distances with average linkage. Distances are calculated in the original 384-dimensional embedding space, not in the reduced UMAP coordinates used for topic identification. The principal partition is defined as the two branches immediately below the final merge.

2.4 Uncertainty and sensitivity analysis

We assess uncertainty using 500 bootstrap repetitions with random seed 42. Debate dates are sampled with replacement within each period. All speeches from a selected date are retained, with the same date multiplicity applied across parties. Profiles and hierarchical clusters are recomputed in each repetition.

For each party pair, we estimate the change in distance between 2018–2019 and 2021–2022 and report the 2.5th and 97.5th bootstrap percentiles. These intervals are pointwise and are not adjusted for multiple comparisons. Clustering stability is measured by the frequency with which the observed principal partition recurs across the 500 repetitions. These frequencies are resampling stability measures, not probabilities that the partitions are correct.

We repeat the distance and clustering calculations using the extended health selection. This checks sensitivity to adding end-of-life discourse; the bootstrap is not repeated for that selection. All uncertainty assessments remain conditional on the source labels, speech selection, and embedding model, and do not explicitly account for dependence across debate dates.

We also assess sensitivity to speaker weighting. Within each party-period group, we first average speech embeddings for each speaker, then average these speaker means with equal weight per speaker. Speaker means are not individually normalized; normalization is applied to the resulting party-period profile. This reduces the influence of speakers who contribute more interventions. The bootstrap is not repeated under this alternative weighting.

3 Results

3.1 Sample composition

The main sample contains 2,400 speeches, with substantial variation across party categories and periods (Table 1). FdI and Lega contribute only 44 and 43 speeches respectively in 2018–2019, whereas FdI contributes 570 in 2021–2022. These differences motivate the uncertainty assessment and caution against interpreting changes in group profiles independently of sample composition.

3.2 Changing semantic groupings

Figure 1 shows the hierarchical clustering of party profiles in each period. The principal partitions change across the three periods, but differ in their stability.

In 2018–2019, FI and PD are the closest pair, with a cosine distance of 0.0061. They join FdI in one principal branch, while Lega and M5S form the other. This partition is reproduced in

Table 1: Number of speeches in the main analytical sample.

Category	2018–2019	2020	2021–2022	Total
PD	127	150	224	501
FdI	44	114	570	728
M5S	188	170	179	537
FI	81	103	169	353
Lega	43	94	144	281
Total	483	631	1,286	2,400

only 40.4% of bootstrap samples, indicating that the early-period grouping is sensitive to the debate dates sampled.

In 2020, the principal split separates PD and M5S from FI, Lega, and FdI. The PD–M5S distance is 0.0082, compared with 0.0292 in 2018–2019. FI and Lega are also close, with a distance of 0.0074. The observed principal partition recurs in 96.4% of bootstrap samples.

In 2021–2022, FdI forms a separate principal branch, while PD, M5S, FI, and Lega belong to the other branch. Distances among these four categories range from 0.0039 to 0.0098, whereas their distances to FdI range from 0.0253 to 0.0458. The observed principal partition recurs in 99.8% of bootstrap samples (Table 2).

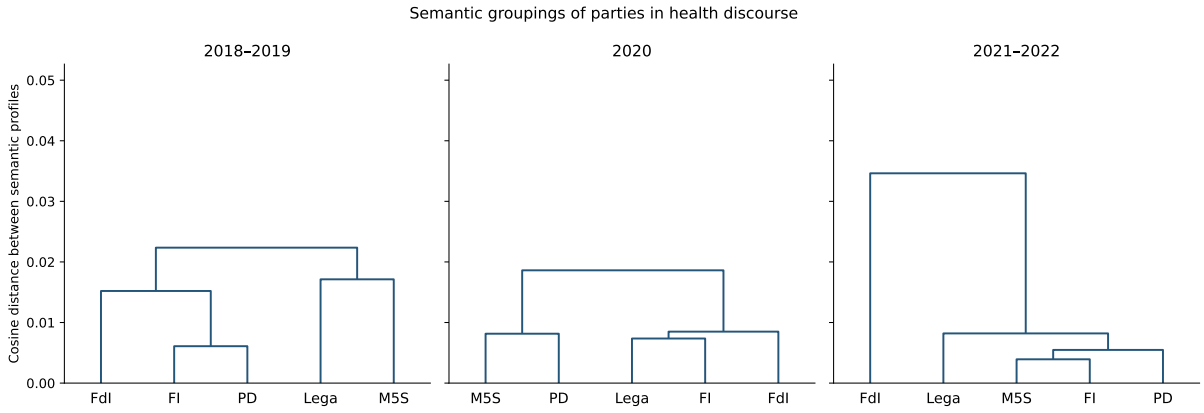


Figure 1: Hierarchical clustering of party-period profiles using cosine distance and average linkage. The vertical scale is shared across periods. Horizontal ordering has no ideological interpretation.

Table 2: Principal partitions and debate-date bootstrap stability.

Period	Principal partition	Bootstrap frequency
2018–2019	FI, FdI, PD / Lega, M5S	40.4%
2020	FI, FdI, Lega / M5S, PD	96.4%
2021–2022	FI, Lega, M5S, PD / FdI	99.8%

3.3 Distance changes and temporal differences

Figure 2 displays the distance changes and their bootstrap intervals. Comparing 2021–2022 with 2018–2019, five party pairs have distance-change intervals entirely below zero: PD–M5S, PD–Lega, M5S–FI, M5S–Lega, and FI–Lega. For example, the PD–M5S distance decreases by 0.0235, with a percentile interval of $[-0.0414, -0.0150]$. The M5S–FI distance decreases by 0.0201, with an interval of $[-0.0406, -0.0100]$.

All four distances involving FdI increase between the first and final periods, but their bootstrap intervals cross zero. The stability of FdI's separate final-period branch should therefore

be distinguished from evidence about changes in individual pairwise distances. A stable partition within 2021–2022 does not establish that each increase relative to 2018–2019 is distinguishable from resampling variation. The reported intervals are also unadjusted for the ten pairwise comparisons.

Within-category comparisons show that all five profiles differ between the first and final periods. FdI has the largest observed temporal distance, 0.0633, followed by FI at 0.0431, PD at 0.0392, M5S at 0.0358, and Lega at 0.0313. These are descriptive rankings: uncertainty intervals for within-category temporal distances were not calculated.

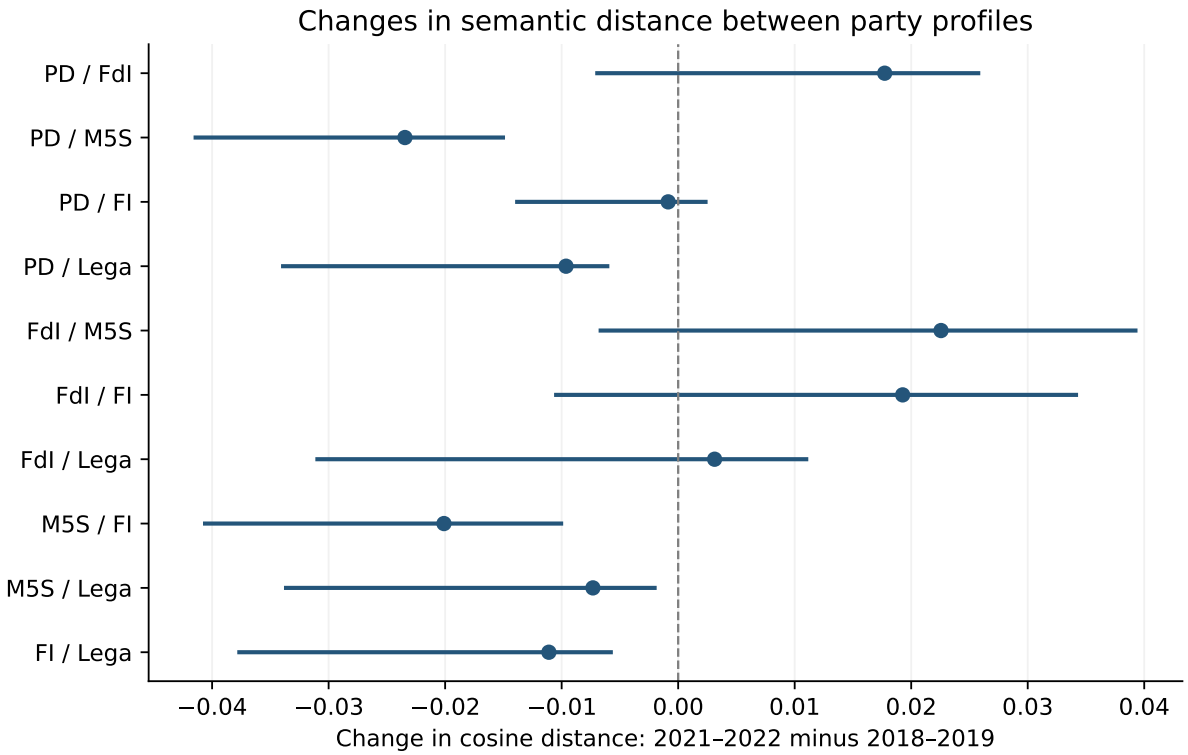


Figure 2: Changes in between-party cosine distances from 2018–2019 to 2021–2022. Points indicate observed changes; horizontal lines show pointwise 95% percentile intervals from 500 debate-date bootstrap repetitions. Negative values indicate closer profiles in the final period. Intervals are not adjusted for multiple comparisons.

3.4 Sensitivity to the health selection

The main analysis includes only speeches assigned to BERTopic topic 0. As a sensitivity check, we also include topic 14, concerning end-of-life discourse. This adds 345 speeches across the five party categories, increasing the sample from 2,400 to 2,745 interventions. The principal partition remains unchanged in all three periods, preserving the broad grouping pattern.

However, several pairwise distances change substantially. In 2021–2022, the M5S–Lega distance increases from 0.0098 to 0.0297, while the FdI–FI distance decreases from 0.0340 to 0.0201. Thus, the persistence of the principal partitions does not imply that the distance estimates are invariant to the definition of health discourse.

The sensitivity analysis supports the persistence of the root partitions under one broader selection. It does not validate the health classification or establish the stability of every relationship within the clustering hierarchy.

3.5 Sensitivity to speaker weighting

Giving equal weight to each speaker preserves the principal partitions observed in 2020 and 2021–2022. The PD–M5S versus FI–Lega–FdI split in 2020 and the separate FdI branch in 2021–2022 therefore persist under this alternative weighting.

The 2018–2019 partition changes: M5S forms a separate branch, while FI, FdI, Lega, and PD join the other branch. This sensitivity is consistent with the low bootstrap frequency of the original early-period partition.

The check supports the persistence of the later-period principal partitions under equal speaker weighting. It does not establish that pairwise distances or lower-level branches are unchanged, and bootstrap uncertainty is not re-estimated under this weighting.

4 Discussion and Conclusion

The analysis identifies a changing organization of semantic similarities across the five source party categories. The principal partition observed in 2018–2019 is sensitive both to debate-date resampling and to speaker weighting. By contrast, the partitions observed in 2020 and 2021–2022 persist under both the extended health selection and equal speaker weighting.

The main finding is therefore a relatively stable pattern within the two pandemic periods, rather than a firmly established transition from a single pre-pandemic alignment. In 2020, PD and M5S form one principal branch, while FI, Lega, and FdI form the other. In 2021–2022, FdI forms a separate branch from the remaining four categories. Five pairwise first-to-last-period distance changes have pointwise bootstrap intervals entirely below zero. However, the observed increases involving FdI have intervals crossing zero. Stability of the final-period partition and uncertainty about individual distance changes are distinct findings.

These patterns concern average semantic content. They should not be interpreted as direct evidence of political agreement or ideological convergence. Parties can discuss the same policies while defending opposing positions, and contextual embeddings may capture shared subjects alongside differences in framing and style. Likewise, changes in the issues discussed, participating speakers, and types of interventions can alter the profiles.

Several limitations qualify the findings. Health-related speeches are selected through an exploratory topic model without a validated estimate of classification accuracy. The extended selection preserves the principal partitions but changes some distances substantially. Party labels are inherited from ItaParlCorpus and may not reflect affiliations at the date of each speech. The samples are unequal across categories and periods, and the bootstrap does not explicitly account for dependence across dates or uncertainty in the embedding model and speech selection. The intervals are also unadjusted for multiple comparisons.

Within these limits, the report provides a reproducible description of how semantic groupings in parliamentary health discourse differ before and during the pandemic. An extension focused on political positions would require explicit identification and validation of policy stances, rather than interpreting semantic proximity alone as agreement.

Data and Code Availability

The accompanying GitHub repository contains the corpus preparation and analysis notebooks, saved speech embeddings, and outputs underlying this report. The workflow separates data preparation, health-speech identification, and semantic comparisons. Run metadata records input checksums, software versions, and analysis settings.

Acknowledgements

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The present extension rebuilds the preparation and analysis workflow and shifts the focus from lexical comparisons between political blocs to semantic comparisons across individual party categories. The additional analyses and interpretations presented here are the responsibility of the author.

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